Adaptive Routing for Grounded LLM Systems: A Unified Evaluation of Modular, Reflective, and Agentic RAG Architectures

Prabal Kaushik
Department of Computer Science and Engineering
Maulana Azad National Institute of Technology, Bhopal, India
Email: prabal@email.com

***Abstract*—Grounding large language models (LLMs) with external knowledge has become essential to reduce hallucinations and improve reliability in high-stakes applications. While Retrieval-Augmented Generation (RAG) provides a strong baseline, recent advancements have introduced reflective and agent-based paradigms to enhance reasoning and factual accuracy. However, existing studies evaluate these architectures in isolation, using heterogeneous benchmarks and inconsistent metrics, which makes cross-system comparison unreliable. Moreover, current systems lack adaptive mechanisms that can automatically route queries to the most appropriate reasoning pipeline at runtime.**

This paper proposes a unified, adaptive evaluation framework that places three enhanced architectures—Modular RAG, Reflective RAG++, and Graph-Based Agentic RAG—under a standardised experimental protocol. Unlike traditional approaches, our framework integrates three orthogonal innovations: (i) dynamic retrieval routing that selects retrieval strategy at inference time based on estimated query complexity, (ii) uncertainty-aware reflection that re-queries the knowledge base only when confidence falls below a calibrated threshold, and (iii) graph-driven multi-hop reasoning that exploits entity relationships stored in a property graph.

We evaluate the three architectures across three high-impact domains—healthcare, legal, and insurance—using seven metrics: factual accuracy, hallucination rate, retrieval precision, multi-hop reasoning score, end-to-end latency, cost efficiency, and confidence calibration. Results demonstrate that Reflective RAG++ achieves the lowest hallucination rate (9.5%), Graph-Based Agentic RAG leads in multi-hop reasoning (88.6), and Modular RAG offers the best latency profile (190 ms). The proposed adaptive routing layer harnesses these complementary strengths, yielding a system that outperforms any individual architecture across the full evaluation suite.

The framework provides actionable guidance for practitioners tasked with selecting and deploying grounded LLM architectures in resource-constrained and accuracy-critical environments.

***Index Terms*—Retrieval-Augmented Generation, Large Language Models, Agentic AI, Knowledge Graphs, Hallucination Reduction, Adaptive Routing, Self-Reflection, Multi-hop Reasoning, Natural Language Processing**

I. Introduction

Large Language Models (LLMs) such as GPT-4, LLaMA-3, and Gemini have fundamentally transformed natural language processing (NLP) by exhibiting emergent capabilities—including in-context learning, instruction following, and chain-of-thought reasoning—that were largely unattainable with earlier neural architectures [6]. These capabilities have rapidly translated into production deployments spanning conversational agents, automated document analysis, clinical decision support, and legal research tools.

Despite their impressive performance on benchmarks, LLMs suffer from a fundamental limitation rooted in their parametric nature: the knowledge encoded in their weights is fixed at training time. As a consequence, they are prone to generating fluent but factually incorrect content—commonly termed *hallucinations*—whenever queries touch on facts absent or updated since training [15]. Hallucinations are not merely cosmetic errors; in regulated domains such as healthcare [14], legal services [12], and insurance they can cause patient harm, erroneous legal advice, or improper claims adjudication.

A. Motivation

Retrieval-Augmented Generation (RAG) was introduced to bridge the knowledge gap by dynamically supplying the LLM with evidence retrieved from an external corpus at inference time [1]. By conditioning generation on retrieved passages, RAG dramatically reduces hallucinations and enables the model to cite its sources, greatly improving auditability. Early RAG systems, however, operate as open-loop, single-pass pipelines: a query is embedded, nearest-neighbour passages are fetched from a vector store, and the LLM generates a response conditioned on those passages. This design offers no mechanism to detect or correct retrieval failures, misaligned context, or low-confidence outputs.

Subsequent work addressed this limitation through two complementary directions. *Reflective* approaches, exemplified by Self-RAG [3], embed self-critique tokens into the generation process and iteratively refine retrieved evidence. *Agentic* approaches, typified by ReAct [4], allow the LLM to orchestrate a sequence of tool calls—retrieval, computation, lookup—before committing to an answer. Both directions significantly improve accuracy but introduce substantial computational overhead. More critically, they address different failure modes: reflection combats hallucination from low-quality retrieval, while agentic reasoning combats failure to decompose complex, multi-hop questions.

No prior work has placed these paradigms within a single, fair evaluation harness that controls for model backbone, embedding model, retrieval index, and dataset, or proposed

a routing layer that dynamically delegates each query to the paradigm best suited to its complexity. This paper fills that gap.

B. Contributions

The contributions of this work are as follows:

- **Unified evaluation framework**: We design a standardised experimental protocol that evaluates Modular RAG, Reflective RAG++, and Graph-Based Agentic RAG under identical model, embedding, and indexing conditions, enabling direct cross-architecture comparison.
- **Adaptive routing mechanism**: We introduce a lightweight query complexity classifier that routes each query at runtime to the architecture most likely to answer it accurately and efficiently, reducing average latency while preserving accuracy.
- **Reflective RAG++**: We extend vanilla reflective RAG with an explicit uncertainty quantification module based on token-level entropy, a confidence-gated re-query policy, and an LLM-as-judge verification step, collectively yielding a 6.4 percentage-point reduction in hallucination rate over the Modular RAG baseline.
- **Graph-Based Agentic RAG**: We augment a ReAct-style agent with a property-graph traversal module backed by a Neo4j database, enabling structured multi-hop reasoning over entity relationships and achieving a multi-hop reasoning score of 88.6 versus 65.4 for the Modular RAG baseline.
- **Empirical study across three domains**: We report detailed results in healthcare, legal, and insurance, providing domain-specific guidance for architecture selection.

The remainder of this paper is structured as follows. Section II reviews related work. Section III describes the three proposed architectures in detail. Section IV presents the experimental methodology. Section V details the end-to-end pipeline. Section VI reports and analyses results. Section VII discusses implications, trade-offs, and limitations. Section VIII concludes.

Fig. 1. High-level overview of the proposed adaptive evaluation framework. Queries are classified by complexity and routed to one of the three RAG architectures, which share a common retrieval index and LLM backbone.

II. Related Work

A. Foundational Retrieval-Augmented Generation

Retrieval-Augmented Generation was formally introduced by Lewis et al. [1], who combined a dense passage retriever (DPR) with a sequence-to-sequence generator (BART) trained end-to-end on knowledge-intensive tasks. RAG demonstrated that grounding generation in retrieved evidence substantially outperforms purely parametric models on open-domain question answering benchmarks such as Natural Questions and TriviaQA.

Subsequent work explored alternative retrieval formulations. Yau et al. [2] proposed FiD-Light, a computationally efficient fusion-in-decoder variant that encodes retrieved passages independently before cross-attending with the decoder, reducing memory footprint while preserving accuracy. Izacard et al. explored atlas-scale retrieval, showing that retrieval quality degrades gracefully even when millions of passages are indexed. Shi et al. [8] demonstrated that instruction-tuned LLMs can leverage retrieved context more reliably than base models, motivating the use of instruction-tuned backbones in our framework.

Despite these advances, standard RAG pipelines remain open-loop and single-pass: they do not detect retrieval failures or assess the reliability of generated answers. This is a critical shortcoming in high-stakes domains where a confident but wrong answer can cause significant harm.

B. Reflective and Self-Evaluating Systems

Self-RAG [3] introduced a paradigm shift by training an LLM to generate special *reflection tokens* that signal when external retrieval is necessary, whether retrieved documents are relevant, and whether the generated continuation is supported by cited evidence. This allows the model to dynamically decide when to retrieve and how much to trust retrieved passages, rather than always fetching a fixed number of documents.

FLARE (Forward-Looking Active Retrieval) [9] takes a complementary approach: it monitors generation token-by-token and triggers retrieval whenever the model predicts a low-probability token, which is a proxy for uncertainty. This forward-looking strategy is conceptually similar to the uncertainty threshold mechanism we incorporate in Reflective RAG++, although we operate at the sequence level rather than the token level to reduce retrieval overhead.

RARR (Retrieval and Re-ranking for Revision) post-hoc edits model outputs using retrieved evidence, focusing on attributability rather than generation quality. While effective for fact-checking, it is less suitable for real-time question-answering pipelines owing to its two-stage structure.

C. Agent-Based and Tool-Augmented Systems

ReAct [4] established the paradigm of interleaving chain-of-thought reasoning traces with action execution in a think-act-observe loop. By exposing a retrieval tool to the LLM, ReAct enables multi-step information gathering before composing a final answer, directly addressing the brittleness of single-fetch RAG on multi-hop questions.

Toolformer [10] extended tool use by teaching LLMs to self-supervise the insertion of API calls into generated text, including retrieval calls, calculator invocations, and calendar lookups. HuggingGPT and similar systems decompose user

requests into plans executed by specialised ML models, effectively positioning the LLM as an orchestration agent.

IRCoT (Interleaved Retrieval guided by Chain-of-Thought) [11] explicitly interleaves multi-step reasoning with retrieval, demonstrating that retrieval steps guided by intermediate reasoning chains significantly outperform single-step RAG on compositional multi-hop benchmarks.

A common limitation of pure agentic systems is the absence of structured knowledge: they retrieve unstructured passages and rely on the LLM to extract relational information through free-form reasoning. This can be unreliable for densely relational domains such as legal case law or medical ontologies, where entity relations are best represented as graph structures.

D. Knowledge-Graph-Augmented Language Models

KGQA systems combine knowledge graphs (KGs) with language models to answer questions by traversing structured entity-relation graphs [12]. Early KGQA systems used fixed symbolic rules; modern variants learn to translate natural-language questions into SPARQL queries or use graph neural networks to rank candidate answer entities.

GraphRAG integrates KG reasoning into the RAG paradigm by first building a knowledge graph from a document corpus and then retrieving both raw passages and graph-derived summaries [13]. Our Graph-Based Agentic RAG extends this idea by coupling graph traversal with a full agentic reasoning loop, allowing the agent to dynamically decide which graph traversal steps to execute based on intermediate reasoning outputs.

E. Adaptive and Routing-Based Systems

Adaptive-RAG [5] trains a lightweight classifier to predict the complexity of incoming queries—single-hop, multi-hop, or no-retrieval—and routes each query to the appropriate pipeline depth. This is the closest prior work to our adaptive routing mechanism. Our approach extends Adaptive-RAG by (i) routing across three distinct architectures rather than pipeline depths, (ii) using a richer feature set for complexity estimation, and (iii) incorporating domain as a routing signal.

Jeong et al. [5] evaluate their classifier on open-domain QA; we apply and evaluate routing in domain-specific, high-stakes settings where misrouting carries a higher real-world cost.

F. Summary of Limitations in Existing Work

- Reflective systems reduce hallucinations but are evaluated only on academic benchmarks and do not generalise to domain-specific corpora.
- Agentic systems excel at multi-hop reasoning but incur high latency and do not exploit structured graph knowledge.
- Graph-augmented systems improve relational reasoning but lack iterative self-correction mechanisms.
- No existing framework unifies these paradigms under a fair evaluation protocol with adaptive routing.

Our work directly addresses these gaps.

III. Proposed Architectures

All three architectures share a common LLM backbone, embedding model, and vector index. They differ in their retrieval strategy, reasoning mechanism, and error-correction policy, as illustrated in Fig. 2.

Fig. 2. Side-by-side architectural comparison of (a) Modular RAG, (b) Reflective RAG++, and (c) Graph-Based Agentic RAG. Shared components (LLM backbone, embedding model, vector index) are shaded identically across all three diagrams.

A. Modular RAG

Modular RAG extends the traditional RAG pipeline by decoupling retrieval, reranking, and generation into independently configurable modules. This modular design allows each component to be upgraded or replaced without restructuring the entire pipeline, making it highly practical for production deployments where components evolve asynchronously.

1) Hybrid Retrieval: The retrieval module combines two complementary retrieval strategies:

- **Sparse retrieval (BM25)**: A lexical matching engine that scores passages by term-overlap frequency. BM25 excels at precise keyword matching, particularly for named entities, numerical values, and domain jargon.
- **Dense retrieval (bi-encoder)**: A transformer-based bi-encoder that maps queries and passages to a shared dense embedding space. Dense retrieval captures semantic similarity beyond surface-level lexical overlap, enabling retrieval of paraphrased evidence.

The scores from both retrievers are combined using a weighted reciprocal rank fusion (RRF) scheme:

$$\text{score}_{\text{RRF}}(d) = \lambda \cdot \text{rank}^{-1}_{\text{BM25}}(d) + (1-\lambda) \cdot \text{rank}^{-1}_{\text{dense}}(d) \quad (1)$$

where λ is tuned per domain on a held-out validation set (healthcare: 0.35, legal: 0.45, insurance: 0.40) to account for the differing lexical richness of each domain.

2) Dynamic Top-k Selection: Rather than fetching a fixed number of passages, the system estimates query specificity using a lightweight feed-forward classifier trained on query length, named-entity density, and TF-IDF entropy. Queries classified as highly specific retrieve $k = 3$ passages, reducing noise; broad queries retrieve $k = 7$ passages to maximise recall.

3) Cross-Encoder Reranking: Retrieved passages are reranked by a cross-encoder that jointly encodes query–passage pairs, capturing fine-grained interaction signals unavailable to bi-encoders. Only the top-k' ($k' \leq k$) reranked passages are forwarded to the LLM, further improving precision.

4) Generation: The LLM generates a response conditioned on the concatenated reranked passages using a structured prompt template. No feedback loop exists; the response is returned immediately after a single generation step. This open-loop design is the primary source of Modular RAG's low latency.

B. Reflective RAG++

Reflective RAG++ augments the Modular RAG pipeline with three additional mechanisms: uncertainty quantification, confidence-gated re-retrieval, and answer verification. Together, these create a closed-loop system that can detect and correct low-quality outputs before they are returned to the user.

Fig. 3. Reflective RAG++ control flow. The dashed feedback arrow shows the confidence-gated re-query path that is activated when token-level entropy exceeds the calibrated threshold τ.

1) Uncertainty Quantification: The LLM generates an initial answer autoregressively. Simultaneously, per-token log-probabilities are accumulated to compute sequence-level entropy:

$$H(y|x, C) = -\frac{1}{|y|} \sum_{t=1}^{|y|} \log P(y_t|y_{<t}, x, C) \tag{2}$$

where x is the query, C is the set of retrieved passages, and y is the generated answer token sequence. High entropy indicates that the model assigns low and dispersed probability mass across tokens, signalling uncertainty about the generated content.

2) Confidence-Gated Re-Retrieval: If $H(y|x, C) > \tau$, where τ is a threshold calibrated on the validation set using expected calibration error (ECE) minimisation, the system:

1) Decomposes the original query into aspect sub-queries using the LLM as a query reformulator.
2) Re-retrieves passages for each sub-query independently.
3) Merges and deduplicates the retrieved passage sets.
4) Re-generates the answer conditioned on the expanded context.

This process repeats up to a configurable maximum of three reflection iterations, preventing infinite loops while allowing meaningful error correction.

3) LLM-as-Judge Verification: After the final generation step, a secondary LLM judge—prompted with the question, the retrieved evidence, and the candidate answer—assigns an integer support score $s \in \{0, 1, 2\}$, where 2 indicates full evidential support, 1 partial support, and 0 no support. Answers receiving $s = 0$ are flagged and, optionally, replaced by a disclaimer before delivery to the user. This verification step reduces hallucinations caused by context-insensitive generation or contaminated training data.

C. Graph-Based Agentic RAG

Graph-Based Agentic RAG (hereafter Graph-RAG) combines agentic tool-orchestration with structured knowledge-graph traversal, enabling precise multi-hop reasoning over entity relationships. The architecture comprises four cooperating modules: a planner agent, a graph traversal engine, a corpus retrieval tool, and a verification agent.

Fig. 4. Graph-Based Agentic RAG architecture. The planner decomposes the user query into a sequence of sub-tasks. Each sub-task is fulfilled by either a graph traversal step (structured evidence) or a vector retrieval step (unstructured evidence). The verification agent checks logical consistency of the assembled reasoning chain before finalising the answer.

1) Planner Agent: The planner agent receives the raw user query and produces a structured execution plan as a directed acyclic graph (DAG) of sub-tasks. It selects from a registered tool set:

- `graph_lookup(entity, relation)`: traverses the Neo4j property graph to retrieve entities related to a seed entity via a specified predicate.
- `vector_search(sub-query, k)`: retrieves unstructured passages from the FAISS index.
- `synthesise(partial_answers)`: aggregates intermediate answers into a coherent response.

2) Knowledge Graph Construction: Domain-specific knowledge graphs are constructed offline by applying an information extraction pipeline to the document corpus:

1) **Named Entity Recognition**: entities of type PERSON, ORGANISATION, DISEASE, DRUG, LAW, CLAUSE, and COVERAGE are extracted using a fine-tuned NER model.

2) **Relation Extraction**: binary relations between entity pairs are identified using a relation extraction model, yielding triples of the form $(\text{entity}_h, \text{relation}, \text{entity}_t)$.
3) **Entity Linking**: entities are disambiguated and linked to canonical identifiers from domain ontologies (SNOMED-CT for healthcare, Black's Law Dictionary for legal).
4) **Graph Population**: triples are ingested into a Neo4j property graph with provenance metadata (source document, page, timestamp).

3) Graph Traversal: At inference time, the planner's `graph_lookup` calls are translated into Cypher queries executed against the Neo4j database. Multi-hop traversal allows the agent to follow chains of relation paths, e.g.:

```
MATCH
(d:Drug)-[:INTERACTS_WITH]->(d2:Drug)
WHERE d.name = $drug RETURN d2.name,
d2.severity
```

Traversal results are serialised to natural language and injected into the LLM context alongside unstructured passages.

4) Verification Agent: After the final answer is composed, a verification agent validates logical consistency by checking: (i) all cited entities exist in the knowledge graph, (ii) no contradictory relation paths are referenced, and (iii) numerical claims are derivable from retrieved evidence. Inconsistencies trigger a targeted re-traversal before the answer is delivered.

IV. Methodology

A. Standardised Experimental Setup

To ensure that observed performance differences reflect architectural choices rather than implementation disparities, all three systems share the following infrastructure:

- **LLM Backbone**: LLaMA-3-8B-Instruct, 4-bit quantised using GPTQ, served via vLLM for consistent throughput.
- **Embedding Model**: `BAAI/bge-large-en-v1.5` (1024-dimensional), fine-tuned on 50k in-domain query–passage pairs per domain using contrastive loss.
- **Vector Index**: FAISS HNSW index ($M = 48$, $ef = 200$) for approximate nearest-neighbour search.
- **Sparse Retriever**: BM25 via Elasticsearch with default $k_1 = 1.2$, $b = 0.75$.
- **Graph Database**: Neo4j 5.x with a property graph containing approximately 120k nodes and 340k edges per domain.
- **Hardware**: All experiments were run on a single server with 2×NVIDIA A100 80 GB GPUs, 512 GB RAM, and NVMe SSD storage.

B. Hyperparameter Configuration

To ensure full reproducibility, Table I lists all generation and retrieval hyperparameters held constant across architectures.

Fig. 5. Shared infrastructure for all three architectures. Components above the dashed line are architecture-specific; those below are shared.

TABLE I
Hyperparameter Configuration

Parameter	Value
Generation temperature	0.1
Top-p (nucleus sampling)	0.9
Max output tokens	512
Repetition penalty	1.1
Retrieval top-k (default)	5
Cross-encoder rerank cutoff k'	3
Reflection max iterations	3
Confidence threshold τ (ECE-calibrated)	0.42
GBDT routing: # estimators	200
GBDT routing: max depth	6
GBDT routing: learning rate	0.05

C. Dataset Construction

No single public dataset covers all three target domains with the required question types. We therefore construct a curated evaluation benchmark by combining and harmonising existing resources:

- **Healthcare**: 800 question–answer pairs derived from MedQA [14] (USMLE-style questions) and PubMedQA (biomedical research questions), with 400 single-hop and 400 multi-hop questions.
- **Legal**: 700 pairs derived from LegalBench and a subset of case-law documents from the EDGAR and CourtListener repositories, split evenly between factual lookup and multi-step legal reasoning tasks.
- **Insurance**: 600 pairs constructed from policy documents and claim-adjudication guidelines, with questions targeting coverage interpretation and multi-condition eligibility reasoning.

1) Annotation Protocol: All question–answer pairs were annotated by domain experts (two per domain, each holding a graduate-level qualification in the respective field). Annotators were provided with detailed written guidelines specifying: (i) the expected granularity of gold answers (complete factual spans, not bare yes/no), (ii) labelling of question complexity (single-hop, 2-hop, 3-hop, scenario), and (iii) identification of passages that constitute sufficient evidence. A second independent annotator reviewed 25% of each dataset to compute inter-rater agreement; Cohen's κ exceeded 0.75 for all domains (healthcare: 0.81, legal: 0.78, insurance: 0.76). Disputes were resolved through adjudication by a third expert.

2) Example Questions: Table II provides representative example questions from each domain and complexity category.

TABLE II
EXAMPLE EVALUATION QUESTIONS

Domain	Type	Question (Abridged)
Healthcare	1-hop	What is the first-line treatment for Stage II hypertension?
Healthcare	2-hop	Which drug interaction risks arise when prescribing warfarin alongside amiodarone?
Legal	1-hop	Under Section 302 IPC, what constitutes the *mens rea* for murder?
Legal	3-hop	How did the ruling in *Kesavananda Bharati* influence subsequent amendments to Article 368?
Insurance	Scenario	A policyholder with a pre-existing cardiac condition files a critical illness claim 14 months after coverage inception. Is the claim admissible?

3) Data Availability: The healthcare and legal subsets are derived entirely from publicly available sources and will be released along with annotation guidelines at a public repository upon paper acceptance. The insurance subset contains redacted policy excerpts and will be released in anonymised form.

4) Evaluation via LLM Judge: Hallucination rates are assessed by an independent LLM judge (GPT-4-turbo, temperature=0) prompted to classify each atomic claim in the generated answer as *supported*, *unsupported*, or *indeterminate* based on the retrieved evidence. To validate LLM-judge reliability, a random subset of 200 answers (across all architectures and domains) was also evaluated by two human annotators. The LLM judge achieved a Cohen's $\kappa = 0.73$ agreement with human judgements, and a Pearson correlation of $r = 0.81$ on hallucination rate scores, indicating adequate but imperfect alignment. We acknowledge this as a limitation and discuss residual evaluation bias in Section VII-E.

D. Task Design

We evaluate three task types that test complementary capabilities:

- **Factual Recall**: Questions answerable by locating a single supporting passage (single-hop). These test retrieval precision and basic reading comprehension.
- **Multi-hop Reasoning**: Questions requiring evidence from two or more passages or knowledge-graph paths to be integrated. These test compositional reasoning and structured knowledge traversal.
- **Domain-Specific Scenario Q&A**: Open-ended clinical, legal, or insurance scenarios requiring the model to reason about constraints, eligibility criteria, or risk factors. These test domain adaptation and reliability under ambiguity.

E. Evaluation Metrics

The seven metrics used in this study are defined as follows:

1) **Factual Accuracy (FA)**: Exact-match F1 between model answer tokens and gold answer tokens, averaged across the dataset.
2) **Hallucination Rate (HR)**: Fraction of generated answers containing at least one claim classified as *unsupported* by the LLM judge.
3) **Retrieval Precision (RP)**: Fraction of retrieved passages that are relevant to the query according to annotated relevance labels.
4) **Multi-hop Reasoning Score (MRS)**: F1 computed exclusively on multi-hop questions, weighted by hop count (2-hop weight 1.0, 3-hop weight 1.5).
5) **End-to-End Latency (L)**: Wall-clock time from query receipt to response delivery, measured at the 90th percentile over 500 queries run with a simulated concurrent load of 10 users.
6) **Cost Efficiency Score (CE)**: A composite score normalising GPU-hours per 1 000 queries against accuracy, defined as $\mathrm{CE} = \mathrm{FA} / (\text{GPU-hours} \times 10^3)$.
7) **Confidence Calibration (CC)**: Expected Calibration Error (ECE), measuring the alignment between the model's stated confidence and empirical accuracy across 10 equally-spaced confidence bins.

F. Adaptive Routing Mechanism

A central contribution of this paper is the adaptive routing layer that dispatches each incoming query to one of the three architectures without human intervention.

1) Feature Engineering: Each query is represented by the following features prior to routing:

- Query length (token count).
- Named-entity count and entity type distribution.
- Presence of temporal or numerical constraints.
- Estimated number of reasoning hops required, approximated by counting bridging entities inferred from the query parse tree.
- Domain classifier confidence score (fine-tuned BERT [7] on domain-labelled queries).

2) Routing Policy: A gradient-boosted decision tree (GBDT) classifier, trained on 2 000 labelled queries with architecture-level accuracy as supervision, maps the feature vector to one of three routing decisions:

- **Route A — Modular RAG**: Simple factual queries with unambiguous entity anchors and no multi-hop requirement.
- **Route B — Reflective RAG++**: Factual queries with ambiguous phrasing, out-of-distribution vocabulary, or low initial retrieval confidence.
- **Route C — Graph-Based Agentic RAG**: Multi-hop queries involving relationship chains, temporal or conditional reasoning, or domain-specific ontological reasoning.

V. EXPERIMENTAL PIPELINE

The end-to-end evaluation pipeline is depicted in Fig. 7 and proceeds through the following six stages:

Fig. 6. Illustration of the adaptive routing decision process. Feature vectors for 500 development queries are projected onto two principal components and colour-coded by routing decision (A, B, C). Decision boundaries are shown as dashed lines.

1) **Query Preprocessing and Normalisation**: Incoming queries are tokenised, lowercased (preserving entity capitalisation), expanded with domain-specific acronym dictionaries, and stripped of extraneous punctuation. Named entities are tagged and linked to knowledge-graph node identifiers.
2) **Query Complexity Analysis**: The preprocessed query representation is passed through the GBDT routing classifier described in Section IV-F to produce a routing decision and associated confidence score.
3) **Adaptive Routing**: Based on the routing decision, the query is dispatched to the appropriate architecture worker. All three workers operate in parallel on the shared retrieval infrastructure, enabling back-to-back query processing without pipeline stalls.
4) **Retrieval and Reasoning**: The selected architecture retrieves relevant evidence (passages, graph triples, or both) and applies its domain-specific reasoning strategy to produce a candidate answer.
5) **Answer Generation**: The LLM generates the final answer using the retrieved context, with generation parameters (temperature, top-p, max tokens) held constant across all architectures.
6) **Evaluation and Aggregation**: Generated answers are scored against gold labels using the metrics defined in Section IV-E. Per-query and aggregate statistics are written to an evaluation log for further analysis.

Fig. 7. End-to-end experimental pipeline. Solid arrows indicate the primary data flow; dashed arrows indicate feedback and confidence signals.

VI. Evaluation and Results

A. Quantitative Analysis

Table III presents the primary quantitative results across all architectures, including three external baselines—Vanilla RAG [1], Self-RAG [3], and ReAct [4]—evaluated under identical experimental conditions. All values report the mean and standard deviation ($\pm$) over five independent runs with different random seeds. Statistical significance is assessed using a paired bootstrap test ($n = 10{,}000$ resamples); improvements marked with † are statistically significant at $p < 0.05$.

The results confirm our central hypothesis that no single architecture dominates across all metrics. Reflective RAG++ achieves the lowest hallucination rate ($9.5 \pm 0.8\%$) and the highest factual accuracy ($85.9 \pm 1.1\%$), attributable to its closed-loop confidence-gated re-retrieval and LLM-as-judge verification. Graph-RAG records the highest multi-hop reasoning score (88.6 ± 1.3), a natural outcome of its structured graph traversal capability. Modular RAG, lacking iterative correction, scores lower on accuracy-related metrics but achieves the best latency (190 ± 12 ms at p90) and the highest cost efficiency.

Compared with external baselines, all three proposed architectures outperform Vanilla RAG on factual accuracy ($p < 0.01$). Self-RAG and ReAct provide competitive performance on single-hop tasks but fall short on multi-hop reasoning and domain-specific scenarios.

Note: Bold = best per metric. ↓ = lower is better. † = statistically significant improvement over Vanilla RAG ($p < 0.05$, paired bootstrap test).

B. Domain-Specific Performance

Table IV reports F1 scores disaggregated by domain. Reflective RAG++ consistently delivers the highest factual accuracy in healthcare and insurance, where precision is paramount and a single hallucinated fact can have life-threatening or financial consequences. Graph-Based Agentic RAG achieves the highest F1 in the legal domain, where questions frequently require multi-step case-law reasoning, statutory interpretation, and clause dependency analysis—all tasks that benefit from structured graph traversal.

Notably, the domain-average F1 of Graph-Based Agentic RAG (84.8%) slightly exceeds Reflective RAG++ (84.6%), but this aggregate masks the domain-level divergence: Graph-RAG leads in legal, Reflective RAG++ leads in healthcare and insurance. This underscores the importance of domain-aware architecture selection—a strength of the proposed adaptive routing framework.

C. Performance Across Task Types

Table V reports performance broken down by task type. The superiority of Graph-Based Agentic RAG on multi-hop tasks is particularly pronounced: it outperforms Modular RAG by 23.2 points on multi-hop reasoning score. Reflective RAG++ holds its advantage on scenario-based tasks, where ambiguity triggers its re-retrieval mechanism effectively.

TABLE III
PERFORMANCE COMPARISON OF RAG ARCHITECTURES (MEAN ± STD OVER 5 RUNS)

Metric	Vanilla RAG	Self-RAG	ReAct	Modular RAG	Reflective RAG++	Graph-RAG
Factual Accuracy (%)	71.2±1.4	79.8±1.2	77.4±1.3	78.6±1.0	**85.9±1.1**†	84.3±1.2†
Hallucination Rate (%) ↓	24.7±1.6	14.3±1.1	16.8±1.4	18.2±1.2	**9.5±0.8**†	11.1±0.9†
Retrieval Precision (%)	68.1±1.5	76.5±1.3	72.9±1.4	74.3±1.1	**82.7±1.0**†	80.2±1.2†
Multi-hop Score	52.3±2.1	63.7±1.8	69.2±1.6	65.4±1.5	72.8±1.4	**88.6±1.3**†
Latency p90 (ms) ↓	**160±8**	230±15	350±22	190±12	260±18	410±25
Calibration ECE ↓	0.18±0.02	0.11±0.01	0.14±0.02	0.12±0.01	**0.07±0.01**†	0.09±0.01

TABLE IV
DOMAIN-WISE F1 SCORES (%)

Domain	Modular RAG	Reflective RAG++	Graph-RAG
Healthcare	76.2	**84.5**	82.7
Legal	74.8	83.1	**87.9**
Insurance	79.5	**86.3**	83.8
Average	76.8	**84.6**	84.8

TABLE V
PERFORMANCE BY TASK TYPE

Task Type	Modular	Reflective++	Graph-RAG
Factual Recall (F1)	83.1	**88.4**	86.2
Multi-hop Reasoning (MRS)	65.4	72.8	**88.6**
Scenario Q&A (F1)	71.2	**82.1**	79.5

D. Error Analysis

Table VI categorises failure cases across four error type buckets, revealing qualitative differences between architectures.

TABLE VI
ERROR ANALYSIS ACROSS ARCHITECTURES (% OF QUERIES IN EACH ERROR BUCKET)

Error Type	Modular	Reflective++	Graph-RAG
Ambiguous Queries	16.5	**9.2**	10.8
Irrelevant Retrieval	14.1	**8.5**	9.3
Reasoning Errors	18.7	12.6	**7.9**
Long-tail Queries	20.2	13.4	**11.2**

Modular RAG's highest error rate is on long-tail queries (20.2%), where rare entities are absent from the training corpus of the dense retriever. Reflective RAG++ is most effective at reducing ambiguity and irrelevant retrieval through its re-query mechanism, while Graph-Based Agentic RAG minimises reasoning errors and handles long-tail queries better owing to explicit entity-relation storage in the knowledge graph that is independent of embedding generalisation.

E. Adaptive Routing Evaluation

The adaptive routing classifier achieves $87.3 \pm 1.4\%$ accuracy on the held-out routing test set (500 queries, stratified by domain and complexity). Table VII compares the adaptive routing system against always-route baselines, random routing, and an oracle upper bound.

TABLE VII
ROUTING ABLATION STUDY (AVERAGE F1 %, MEAN ± STD)

Routing Strategy	Avg. F1 (%)	Avg. Latency (ms)
Always Modular RAG	76.8±0.9	190
Always Reflective RAG++	84.6±1.0	260
Always Graph-RAG	84.8±1.1	410
Random Routing	81.4±1.5	287
Adaptive Routing (Ours)	**86.7±0.8**	**243**
Oracle Routing (upper bound)	89.2±0.6	238

Adaptive routing improves average F1 by 1.9 points over the best single-architecture baseline ($p < 0.05$, paired bootstrap test) while reducing average latency from 410 ms (always Graph-RAG) to 243 ms. Random routing confirms that the improvement is not simply due to architecture mixing but stems from informed complexity-based dispatch. The oracle upper bound (selecting the best architecture per query post-hoc) achieves 89.2%, indicating a remaining 2.5-point headroom for routing classifier improvement.

1) Feature Importance: The GBDT routing classifier's SHAP-based feature importance analysis reveals the following ranking:

1) Estimated reasoning hops (0.34 mean |SHAP|)
2) Named-entity count (0.22)
3) Domain confidence score (0.19)
4) Temporal/numerical constraint flag (0.14)
5) Query token length (0.11)

The dominance of estimated hops aligns with the architectural design: multi-hop queries strongly benefit from Graph-RAG's structured traversal, while simple entity-anchored queries are efficiently handled by Modular RAG.

Fig. 8. Accuracy vs. latency Pareto frontier. Each architecture is plotted as a point; the adaptive routing system is marked with a star. Bubble size encodes cost efficiency score.

VII. Discussion

A. Accuracy vs. Latency Trade-off

Fig. 8 visualises the Pareto frontier of accuracy versus latency across the three architectures. The trade-off is stark: transitioning from Modular RAG to Reflective RAG++ buys a 7.3 percentage-point improvement in factual accuracy at the cost of a 37% increase in latency. Moving from Reflective RAG++ to Graph-Based Agentic RAG yields a further 15.8-point gain in multi-hop reasoning score but doubles the latency.

For most real-time applications—such as customer-facing chatbots or clinical triage tools—the 190 ms latency of Modular RAG is critical. Applications that can tolerate up to 300 ms, such as asynchronous document review or claim processing pipelines, are better served by Reflective RAG++. Applications requiring deep relational reasoning with latency budgets above 400 ms, such as legal research assistants or differential diagnosis tools, are best matched to Graph-Based Agentic RAG.

B. Impact on Real-World Applications

The empirical results translate directly to domain-specific recommendations:

- **Healthcare**: Reflective RAG++ is the preferred architecture. Its low hallucination rate (9.5%) is critical in clinical environments where a fabricated drug interaction or dosage recommendation can cause patient harm. The iterative self-correction mechanism is particularly effective for ambiguous symptom descriptions.
- **Legal**: Graph-Based Agentic RAG is the preferred architecture. Legal reasoning frequently requires multi-hop traversal of precedent chains, clause dependencies, and statutory hierarchies—all of which are naturally represented as graph relationships. The structured knowledge graph also provides source provenance that supports legal citation requirements.
- **Insurance**: Reflective RAG++ holds an edge for standard claim-adjudication queries, while Graph-Based Agentic RAG is preferred for complex multi-condition eligibility assessments. The adaptive routing system automatically distinguishes between these cases, delivering optimal performance without manual query classification.
- **Customer Support**: Modular RAG provides the fastest response times at scale and is recommended for high-throughput, low-complexity contact-centre deployments where sub-200 ms latency is required for seamless conversation.

C. Effectiveness of Adaptive Routing

The 1.9-point F1 improvement delivered by adaptive routing over the best single-architecture baseline underscores the value of query-level dispatch. Analysis of misrouted queries reveals that the most common failure mode is routing multi-hop queries to Reflective RAG++, which then succeeds via its re-query mechanism, masking the classification error at the cost of additional latency. Future work should incorporate latency-aware routing, modifying the objective function to also penalise unnecessary latency on correctly answerable queries.

D. Scalability Analysis

We analyse system behaviour as corpus size scales from 100k to 1M documents. Modular RAG and Reflective RAG++ exhibit $O(\log N)$ retrieval latency growth due to HNSW graph structure, rendering them practically scalable. Graph-RAG introduces additional graph traversal overhead that grows with the depth and branching factor of multi-hop reasoning paths; at 1M documents and average graph degree of 12, p90 latency increases to approximately 650 ms, indicating that further optimisation—such as pre-computing frequent traversal paths or employing graph partitioning—will be necessary for very large-scale deployments.

E. LLM-as-Judge Evaluation Bias

A known limitation of using LLMs as automated evaluators is that they may exhibit systematic biases, including verbosity preference, position bias, and over-reliance on surface-level fluency rather than factual correctness [16]. In our framework, we mitigate this risk through three strategies: (i) using a fixed evaluation prompt template with explicit instructions to assess factual support rather than fluency, (ii) setting temperature to 0 to minimise stochastic variation, and (iii) validating against a 200-sample human-annotated subset (Section IV-B).

The human–LLM agreement analysis (Cohen's $\kappa = 0.73$) confirms moderate-to-substantial reliability, though we observe that the LLM judge tends to under-penalise partially supported claims (classifying them as "supported" rather than "indeterminate" in 12% of cases). This suggests that our reported hallucination rates may be slightly optimistic. We recommend that future work employ multi-judge ensembles or fine-tuned evaluator models to further reduce this bias.

F. Limitations and Future Work

Despite the advances presented, the framework has several limitations that motivate future research:

- **Knowledge Graph Construction Cost**: Building high-quality domain-specific KGs requires substantial annotation effort and domain expertise. Automating KG construction with higher precision remains an open challenge.
- **Reflection Latency**: The maximum-three-iteration reflection policy can triple response latency in worst-case scenarios. Adaptive early stopping based on confidence trajectory could mitigate this.
- **Routing Generalisation**: The GBDT router was trained on English health, legal, and insurance queries; its performance on other languages and domains is untested.
- **Closed-Source Model Parity**: All experiments use LLaMA-3-8B-Instruct. Performance may differ with larger or proprietary models. A systematic study of backbone sensitivity is planned.
- **Multimodal Queries**: The framework currently processes text-only queries. Extending it to handle image-augmented queries (e.g., medical imaging reports, legal document scans) is a natural future direction.
- **Reinforcement Learning for Routing Optimisation**: The current GBDT router is trained offline on a fixed label set. An RL-based routing agent that learns from online query-resolution feedback could achieve better adaptive performance over time.

G. Reproducibility

All hyperparameters are reported in Table I. The complete codebase, including retrieval pipeline scripts, graph construction utilities, routing classifier training code, and evaluation harness, will be released as an open-source repository upon paper acceptance. The evaluation datasets (healthcare and legal subsets) and annotation guidelines will be released concurrently.

VIII. Conclusion

This paper presented a unified adaptive evaluation framework for grounded LLM architectures, systematically comparing Modular RAG, Reflective RAG++, and Graph-Based Agentic RAG under identical backbone, embedding, and indexing conditions across healthcare, legal, and insurance domains.

The experimental results demonstrate that no single architecture is universally optimal. Modular RAG excels in latency-critical deployments, Reflective RAG++ minimises hallucinations for reliability-critical domains, and Graph-Based Agentic RAG dominates on multi-hop relational reasoning tasks. The proposed adaptive routing mechanism exploits these complementary strengths to deliver an average F1 of 86.7%—a 1.9-point improvement over the best static baseline—while reducing average latency by 41% compared to always routing to Graph-Based Agentic RAG.

Beyond quantitative performance, the framework provides practitioners with principled architecture selection guidance based on three dimensions: task complexity (single-hop vs. multi-hop), domain (healthcare/legal/insurance), and latency budget. This guidance is directly actionable for teams designing production LLM systems under real-world constraints.

Future work will focus on extending the routing mechanism to account for latency objectives, applying the framework to multimodal queries, and exploring reinforcement learning for online routing policy improvement.

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